

STOCHASTIC DATA-DRIVEN DIFFERENTIAL EQUATIONS

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ABSTRACT. This paper presents a brief overview of stochastic data-driven differential equations, a surrogate time-series AI model that combines stochastic differential equations with machine learning.

1. INTRODUCTION

Machine learning models are valued for approximating complex phenomena, but high representational power does not guarantee good generalization. Black-box models like neural networks learn an implicit input-output mapping and can fail silently when inference-time data differs from training data. Since these models typically output only a single predicted value, the output looks the same whether the model is confident or not — nothing warns the user that reliability has dropped.

Underlying this is a more basic issue: the available features are often insufficient to fully explain the target variable. In any real system, some of what determines the outcome is simply not captured by the observed inputs, so a certain amount of residual, unexplained variation is unavoidable. The ‘inductive bias’ built into black-box models — the assumptions about the world embedded in the architecture — implicitly assumes the opposite: that the output is fully determined by the input. This mismatch is what allows failures to go undetected. What’s needed is a model whose structure leaves room for this feature insufficiency rather than assuming it away.

The stochastic differential equation (SDE) is one way to do this, separating a deterministic part (drift) from a random part (diffusion). In the proposed framework, the diffusion term is replaced by an ensemble-based surrogate, σ^{AI} , used not to estimate a “true” diffusion coefficient but as a rough indicator of how well the available features explain the target — it tends to rise when the features are insufficient to explain the phenomenon, or when the input differs from the training data. Choosing an SDE thus reflects an inductive bias that leaves room for this insufficiency rather than assuming it away; a rising σ^{AI} can be read as a sign that reliability may be dropping. Conventional black-box models, which output only a single number, don’t offer this kind of signal. The SDE framework thus offers one way to consider accuracy and reliability together, potentially useful where the available features can’t fully capture the target variable.

2. CONSTRUCTION

The proposed model adopts a stochastic differential equation (SDE) as the basic framework for time-series modeling, with the SDE’s parameters derived in a data-driven manner. As in classical SDE models such as geometric Brownian motion, the drift μ and diffusion σ are not determined by theory alone; they are, in effect, empirical constants that need to be estimated from data. In this sense, applying machine learning to the drift and diffusion terms can be viewed as estimating the

SDE from the time series in a data-driven way, rather than fixing these terms in advance.

Conventional calibration of such empirical constants typically optimizes a single objective: fitting past behavior, after which the estimated value is held fixed and applied uniformly. The proposed framework instead fits $\mu_\theta(x_t)$ and $\sigma_\theta(x_t)$ with two objectives in mind: reproducing observed dynamics, and allowing σ to also serve as a surrogate, σ^{AI} , whose value tends to increase when the observed features are less sufficient to explain the current situation. Because σ^{AI} is re-estimated conditionally rather than fixed, it can reflect, to some extent, whether the present case falls within the range the estimation is based on. This offers one possible way to give the model some indication of its own predictive reliability — a property that fixed constants, deterministic models, and typical black-box approaches do not directly provide.

Theory is used where it can specify a component directly — the SDE formulation, the Euler–Maruyama scheme, and path generation — while the data-driven estimation is applied only to μ and σ , which theory alone does not determine. The resulting ‘hybrid AI model’ combines theory-driven structure with data-driven estimation of the SDE, aiming to retain the interpretability of stochastic process theory while limiting the use of machine learning to the parts where it is needed, and to make some indication of estimation reliability available alongside the prediction.

We now consider a stochastic process $\{S_t\}_{t \in [0, T]}$ satisfying the following stochastic differential equation (SDE):

$$dS_t = S_t(\mu dt + \sigma dW_t), \quad S_0 = S > 0.$$

The explicit solution of the above SDE is given by

$$S_t = S \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right), \quad 0 \leq t \leq T.$$

Therefore, the evolution of S_t is completely characterized by the parameters μ and σ . In practice, however, these quantities are generally unknown and must be estimated from observed data. Various approaches have been proposed for estimating the drift term μ and the diffusion coefficient σ . In this study, we propose a data-driven surrogate approach based on machine learning. Specifically, the drift term μ and the diffusion term σ are learned from a dataset consisting of feature vectors F and corresponding observations S_T within a framework combining deep learning and the Euler–Maruyama approximation. Applying the Euler–Maruyama scheme to a partition of $[0, T]$,

$$0 = t_0 < t_1 < \cdots < t_N = T, \quad \Delta t = t_{i+1} - t_i,$$

we obtain an approximation $\bar{S}_{t_i}$ of S_{t_i} satisfying

$$\bar{S}_{t_{i+1}} - \bar{S}_{t_i} = \bar{S}_{t_i} (\mu \Delta t + \sigma \Delta W_{t_i}).$$

In many practical applications, observations are only available at the initial and terminal times. To reflect this constraint, we adopt the case $N = 1$. Then,

$$\frac{S_T - S}{S} \approx \frac{\bar{S}_T - S}{S},$$

and therefore

$$\frac{S_T - S}{S} \approx \mu T + \sigma \sqrt{T} Z \sim N(\mu T, \sigma^2 T),$$

where $Z \sim N(0, 1)$. Hence, the Euler–Maruyama approximation with $N = 1$ implies that the random variable $\mu T + \sigma \sqrt{T} Z$ provides an approximation to the normalized return $(S_T - S)/S$. The key idea of the proposed method is to replace

this $N = 1$ Euler–Maruyama approximation with deep learning. More specifically, multiple bootstrap datasets

$$\mathcal{D}^1, \dots, \mathcal{D}^K$$

are generated from the original training dataset $\mathcal{D}$ via bootstrap sampling. A deep learning model is independently trained on each bootstrap dataset, yielding approximation functions

$$N_T^1, \dots, N_T^K$$

for the normalized return $(S_T - S)/S$. For a given feature vector F , we define the mean and standard deviation of the ensemble predictions as

$$\mu_T^{AI}(F) = \frac{1}{K} \sum_{i=1}^K N_T^i(F),$$

and

$$\sigma_T^{AI}(F) = \sqrt{\frac{1}{K} \sum_{i=1}^K (N_T^i(F) - \mu_T^{AI}(F))^2}.$$

The quantity $\mu_T^{AI}(F)$ is interpreted as an estimator of the conditional mean of $(S_T - S)/S$ given the feature vector F , while $\sigma_T^{AI}(F)$ serves as an ensemble-based surrogate for the predictive uncertainty associated with this estimate. They serve as data-driven substitutes for the quantities μT and $\sigma\sqrt{T}$ appearing in the Euler–Maruyama approximation:

$$\mu T + \sigma\sqrt{T}Z \longrightarrow \mu_T^{AI}(F) + \sigma_T^{AI}(F)Z.$$

It should be emphasized that $\sigma_T^{AI}(F)$ quantifies the variability of the ensemble predictions under bootstrap resampling and is therefore a surrogate for predictive uncertainty, not a consistent estimator of the conditional standard deviation of $(S_T - S)/S$. In particular, its magnitude depends on the training configuration and the sample size, and it should be interpreted as a relative indicator of informational coverage. When the generated paths are used quantitatively, e.g., for prediction intervals, a calibration step on held-out data is recommended to align the surrogate with empirical coverage. Consequently, for each feature vector F , we obtain the following data-driven stochastic differential equation:

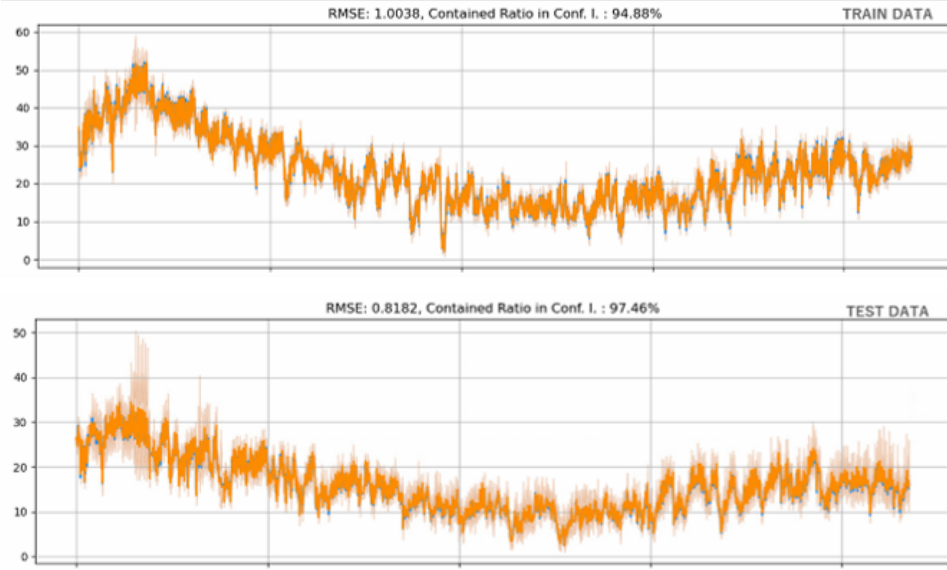
$$dS_t^{AI} = S_t^{AI} \left(\frac{\mu_T^{AI}(F)}{T} dt + \frac{\sigma_T^{AI}(F)}{\sqrt{T}} dW_t \right), \quad S_0^{AI} = S > 0.$$

3. EXPERIMENT

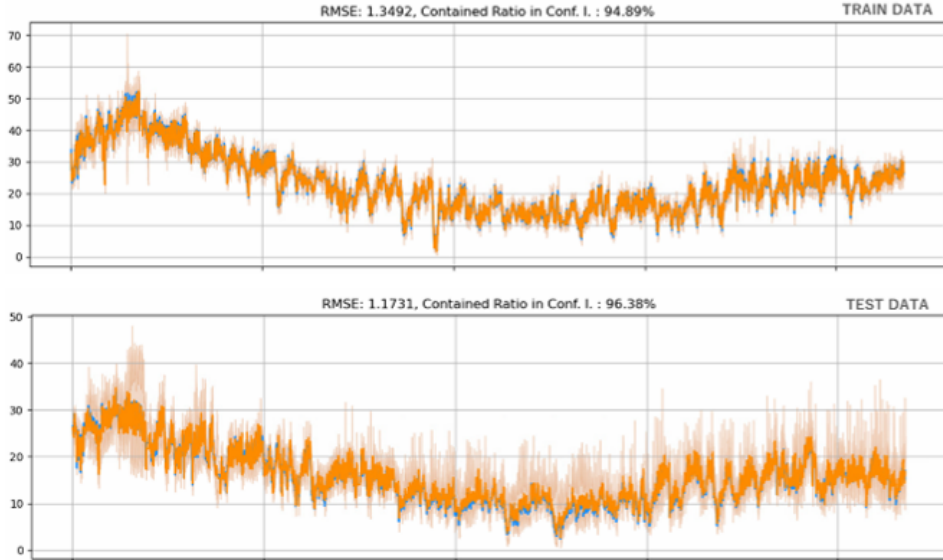
We evaluate the proposed model on the ETT (Electricity Transformer Temperature) dataset. The input features are deliberately restricted to only the current time and the current oil temperature (OT). This minimal state representation is not intended to test the Markov property itself. Rather, it is designed to create a setting in which the available features are intentionally insufficient to fully explain the future evolution of OT, thereby introducing a high degree of uncertainty. This allows us to examine whether the proposed model remains effective even under deliberately feature-limited conditions. Taking the current time as the reference point, we predict OT 1, 2, and 3 hours ahead. We evaluate two aspects simultaneously: (i) predictive accuracy by comparing the performance on the training and test datasets, and (ii) the behavior of the surrogate diffusion term σ^{AI} , specifically whether it increases as the prediction error grows rather than remaining silent. These two aspects are inherently related by the experimental design. The current time and OT alone are unlikely to capture all factors influencing future OT, such

as electrical load and ambient temperature. Consequently, some unexplained variation is expected, and its influence should become more pronounced as the prediction horizon increases. If the proposed model appropriately represents this uncertainty, σ^{AI} should increase together with the prediction error, instead of producing point estimates that fail to indicate declining reliability. Therefore, rather than treating this experiment as a validation of the Markov property itself, we interpret the joint behavior of prediction error and σ^{AI} across prediction horizons as evidence of whether the proposed model can appropriately represent predictive reliability even when the available features are intentionally insufficient.

3.1. 1-hour forecast.



3.2. 2-hour forecast.



3.3. 3-hour forecast.

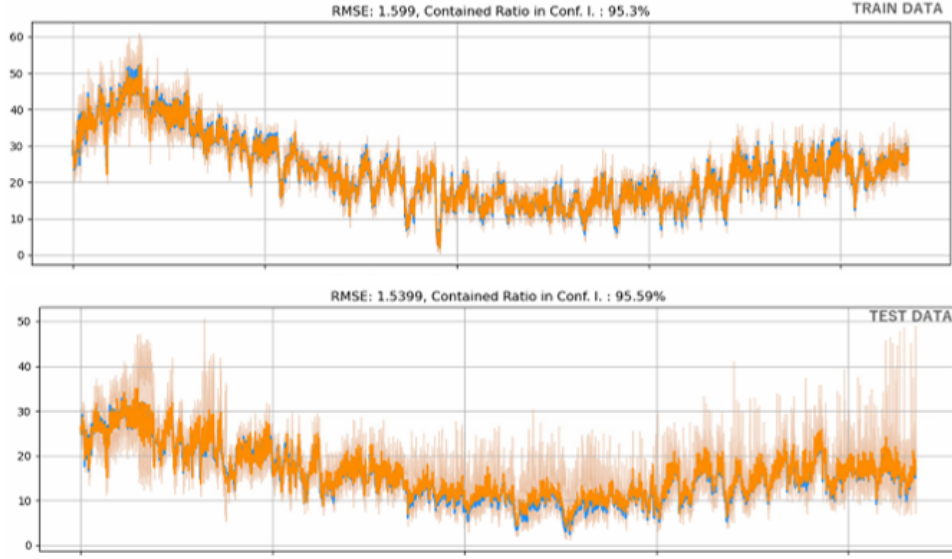


TABLE 1. Experimental results

	Train		Test	
	RMSE	2σ Coverage	RMSE	2σ Coverage
1-hour forecast	1.0038	94.88%	0.8182	97.46%
2-hour forecast	1.3492	94.89%	1.1731	96.38%
3-hour forecast	1.5990	95.30%	1.5399	95.59%

3.4. Results and Discussion. As a result, the discrepancy in RMSE between training and test data remains small across all horizons — for example, from 1.00 to 0.82 at the 1-hour horizon, and from 1.60 to 1.54 even at the 3-hour horizon. The 2σ confidence interval coverage is approximately 94–97% in both cases, consistent with the theoretical value of 95.4%. These results indicate that the proposed model simultaneously satisfies both generalization capability and mathematical consistency.